

For your Amazon Senior Solutions Architect interview, these are exactly the questions I would prepare. Don't memorize long answers; memorize the architecture thinking and a 30–60 second answer for each.

1. How can you make your application scalable for a big traffic day?

I would answer:

"I would first identify the expected traffic, TPS, latency SLA and read/write ratio. Then I would remove single points of failure and make the application horizontally scalable."

AWS architecture:

Users → CloudFront → AWS WAF → Application Load Balancer → Auto Scaling → EC2 / ECS / EKS

↓

ElastiCache RDS/Aurora SQS → Read Replicas

Key techniques:

- CloudFront → absorb/cache static traffic
- WAF → protect application
- ALB → distribute traffic
- EC2 Auto Scaling / ECS / EKS → horizontal scaling
- SQS → decouple burst traffic
- ElastiCache → reduce database load
- RDS Multi-AZ → availability
- Read replicas → read scaling
- DynamoDB → extremely high-scale key/value workloads
- S3 → virtually unlimited object storage
- CloudWatch → monitor and automatically scale

The important Senior Architect statement:

"I would design for elasticity rather than simply provisioning a larger server."

2. How do you achieve DR for your cloud application?

Start with:

"I would first define RPO and RTO because the DR architecture depends on the business requirements."

RPO = How much data can I afford to lose?

RTO = How quickly must I recover?

Primary Region → database replication, S3 replication, backups, configuration/IaC → DR Region

Strategies: Backup & Restore → Pilot Light → Warm Standby → Active/Active

AWS services: Route 53, AWS Backup, S3 Cross-Region Replication, Aurora Global Database, DynamoDB Global Tables, CloudFormation/CDK/Terraform, CloudWatch, Systems Manager.

Strong interview statement:

"DR isn't simply having a backup. I need to demonstrate that I can restore the application, data, configuration and dependencies within the agreed RTO."

3. How do you secure your application on the cloud?

Use defense in depth across network, identity, data and monitoring.

Internet → CloudFront → WAF → ALB → Private Subnets → Application → Private Database

Network: VPC, public/private subnets, Security Groups, NACLs, VPC endpoints, no public database.

Identity: IAM, IAM Roles, least privilege, MFA, temporary credentials, Cognito/OIDC.

Data: KMS, encryption at rest, TLS in transit, Secrets Manager, S3 bucket policies.

Monitoring: CloudTrail, GuardDuty, Security Hub, Config, CloudWatch.

Excellent answer:

"I use defense in depth. I don't depend on a single security control."

4. Describe an architecture you designed

"One architecture I worked on was an enterprise AI platform integrating RAG, agentic workflows and risk technology."

Users → API Gateway / ALB → Authentication → Agent Orchestrator → Risk Agent / RAG Agent / Compliance Agent → OpenSearch V
Ingestion: Documents → S3 → EventBridge → SQS → Processing → OCR / PII / Classification → Chunking → Embeddings → OpenSe
Trade-offs: security, latency, scalability, model cost, RAG quality, observability and multi-tenancy.

5. What was the biggest architectural challenge when designing your cloud application?

Do not say: "The biggest challenge was configuring AWS."

Use this answer:

"The biggest challenge was balancing AI quality, latency, cost and enterprise security."

A higher-quality model can increase cost and latency. More specialized agents can improve outcomes but add LLM calls, cost and latency. Additional security controls can also introduce latency.

"I addressed this by measuring the complete request path, using caching and retrieval optimization, routing simpler tasks to smaller models, limiting agent/tool calls, and applying security controls at the authorization boundary rather than relying on the model."

6. How do you pick one service versus another as a Solutions Architect?

"I start with requirements rather than services."

Business requirements → functional requirements → non-functional requirements → constraints → architecture options → trade

Evaluate scalability, availability, performance, security, cost, operational complexity, integration, team skills, compliance and service maturity.

"If I need a relational transactional database with complex joins, I might choose Aurora/RDS. If I need massive horizontal key-value scaling with predictable access patterns, DynamoDB may be better."

"The requirement drives the service; the service doesn't drive the requirement."

7. Difference between SQL and NoSQL?

SQL: structured schema, tables, relationships, ACID transactions and complex queries. Examples: Aurora, RDS PostgreSQL, RDS MySQL. Good for banking, orders, payments, financial transactions and ERP.

NoSQL: flexible schema, horizontal scaling, access-pattern-driven modeling and high throughput. Examples: DynamoDB, DocumentDB, Neptune and Keyspaces.

SQL example: Customer → Orders → Payments.

DynamoDB example: PK = CUSTOMER#123; SK = ORDER#456.

"NoSQL isn't simply SQL without a schema. The data model and access patterns are fundamentally different."

8. What is cloud computing?

"Cloud computing is the on-demand delivery of computing resources such as compute, storage, databases, networking and AI services over the internet, with elastic capacity and usage-based economics."

Benefits: on-demand, elastic, global, pay-as-you-go, highly available, managed services and automation.

Traditional: buy servers → install → configure → maintain → capacity planning.

Cloud: requirement → provision API → scale automatically → pay for usage.

9. What is AWS Service X?

Use this template: What problem does it solve? How does it work? When would I use it? When would I not use it? What are the alternatives? What are the trade-offs?

Example: Amazon SQS is a managed message queuing service used to decouple distributed components. I would use it when producers and consumers should operate independently, particularly when I need to absorb traffic spikes, handle asynchronous processing or improve resilience.

Producer → SQS → Consumer

Benefits: decoupling, buffering, retry, asynchronous processing and elasticity.

The 9 questions — your mental cheat sheet

Scalability → horizontal scaling + caching + queues + autoscaling

DR → RPO + RT0 + backups + replication + tested recovery

Security → IAM + VPC + encryption + WAF + monitoring

Architecture → requirements → design → trade-offs → implementation

Challenge → quality vs cost vs latency vs security

Service selection → requirements → trade-offs → service

SQL vs NoSQL → transactions/relationships vs scale/access patterns

Cloud → on-demand + elastic + managed + pay-as-you-go

AWS Service X → problem → usage → alternative → trade-off

The interview landscape focuses heavily on trade-offs, cost optimization and design judgment rather than simply memorizing service features. Practice saying: "Given these requirements, I would choose X instead of Y because..."

PART 1: FOUNDATIONS & WELL-ARCHITECTED FRAMEWORK

1. What are the six pillars of the AWS Well-Architected Framework?

Answer: Operational Excellence, Security, Reliability, Performance Efficiency, Cost Optimization and Sustainability. These pillars often trade off against each other; for example, multi-AZ improves reliability but increases cost.

2. What is the difference between Scalability and Elasticity?

Answer: Scalability is the ability to handle increased load, vertically or horizontally. Elasticity is the ability to automatically scale up and down based on demand. AWS is designed to be elastic, whereas on-premise systems are typically only scalable.

3. Explain the AWS Shared Responsibility Model.

Answer: AWS is responsible for security of the cloud, including physical infrastructure and the hypervisor. The customer is responsible for security in the cloud, including IAM, network configuration, OS patching for EC2 and data encryption. The boundary shifts by service.

4. Region vs. Availability Zone vs. Edge Location?

Answer: A Region is a geographic area. An AZ is one or more discrete data centers within a region with redundant power and networking. An Edge Location is a CloudFront Point of Presence used for caching content closer to users.

5. What is Stateful vs. Stateless architecture?

Answer: Stateless systems keep no local session data; any instance can serve any request because state is externalized to a database or cache. Stateful instances hold local state and require session affinity or replication. Modern AWS designs prefer stateless tiers.

6. What are AWS Organizations and Service Control Policies?

Answer: Organizations centrally manage multiple accounts. SCPs are guardrails at the OU or account level defining the maximum permission boundary. SCPs cannot grant permissions; they only restrict what IAM otherwise allows.

7. When would you use AWS Control Tower?

Answer: Use Control Tower to automate a secure multi-account landing zone with baseline guardrails, centralized logging and identity federation. It is useful when managing several AWS accounts.

8. What are the hybrid connectivity options on AWS?

Answer: Site-to-Site VPN provides IPsec over the internet and quick setup. Direct Connect provides dedicated connectivity and predictable performance. Transit Gateway provides a hub for VPCs and on-premises networks. Enterprises often use Direct Connect with VPN as backup.

PART 2: NETWORKING & VPC

9. Walk me through the components of a VPC.

Answer: CIDR block, subnets per AZ, route tables, Internet Gateway, NAT Gateway, Security Groups, NACLs and VPC Endpoints.

10. What is the difference between a Public and Private Subnet?

Answer: A public subnet has a route to an Internet Gateway and instances may have public IPs. A private subnet routes outbound traffic through a NAT Gateway; instances have no public IP and cannot receive unsolicited inbound internet traffic.

11. Security Group vs. NACL?

Answer: A Security Group is stateful, instance-level and has allow rules only. A NACL is stateless, subnet-level and supports allow and deny rules.

12. VPC Peering vs. Transit Gateway?

Answer: VPC Peering is point-to-point and non-transitive. Transit Gateway is hub-and-spoke, supports transitive routing and connects many VPCs, VPNs and Direct Connect gateways. Use TGW for complex networks.

13. What is a VPC Endpoint?

Answer: It provides private connectivity to AWS services without traversing the internet. Gateway endpoints are free for S3/DynamoDB; interface endpoints use PrivateLink and support most other services.

14. NAT Gateway vs. NAT Instance?

Answer: NAT Gateway is managed and highly available within an AZ. NAT Instance is self-managed EC2, cheaper at low volume but requiring manual HA and scaling. Default to NAT Gateway for production.

15. ALB vs. NLB vs. GLB?

Answer: ALB is Layer 7 for HTTP/HTTPS and path-based routing. NLB is Layer 4 for TCP/UDP, ultra-high performance and static IPs. Gateway Load Balancer inserts third-party virtual appliances such as firewalls or IDS.

PART 3: COMPUTE & SCALING

16. How do you choose EC2 instance types?

Answer: M-series general purpose; C-series compute optimized; R/X-series memory optimized; I/D-series storage optimized; P/G/Inf-series GPU or accelerated computing. Use Graviton for price/performance where supported.

17. On-Demand vs. Reserved vs. Savings Plans vs. Spot?

Answer: On-Demand is pay-as-you-go. Reserved Instances and Savings Plans provide discounts for one- to three-year commitments. Spot can offer large discounts but may be interrupted. Mix them for cost optimization.

18. What scaling policies exist for Auto Scaling Groups?

Answer: Target Tracking, Step Scaling, Scheduled Scaling and Predictive Scaling. Combine target tracking with scheduled scaling for predictable spikes.

19. Launch Template vs. Launch Configuration?

Answer: Launch Templates are the modern standard and support versioning, mixed instance types and Spot allocation strategies. Launch Configurations are deprecated.

20. EC2 vs. ECS vs. EKS vs. Fargate vs. Lambda?

Answer: EC2 gives full control. ECS is AWS-native orchestration. EKS is managed Kubernetes. Fargate is serverless containers. Lambda is event-driven functions. Choose according to control, runtime, ecosystem and operational needs.

21. ECS Fargate vs. ECS on EC2?

Answer: Fargate removes server management and charges per task. EC2 is cheaper at sustained high scale and supports GPUs and custom AMIs. Default to Fargate for new workloads unless specific hardware needs exist.

22. What is a Lambda Cold Start and how do you mitigate it?

Answer: It is latency during the first invocation after inactivity. Mitigate with Provisioned Concurrency, smaller deployment packages, faster runtimes or avoiding VPC attachment when it is unnecessary.

PART 4: STORAGE

23. S3 Storage Classes — pick one for each scenario.

Answer: Standard for frequent access; Intelligent-Tiering for unknown patterns; Standard-IA for infrequent access; Glacier Instant Retrieval for archive with millisecond access; Glacier Deep Archive for lowest cost when long retrieval times are acceptable.

24. How do you secure an S3 bucket?

Answer: Block Public Access, least-privilege bucket policies, VPC endpoints, SSE-KMS/S3 encryption, HTTPS enforcement, versioning and Object Lock where ransomware protection is required.

25. EBS Volume Types?

Answer: gp3 is general-purpose SSD with decoupled IOPS; io2 is for high IOPS databases; st1/sc1 are throughput or cold HDD. Default to gp3 for new workloads.

26. EBS vs. EFS vs. FSx?

Answer: EBS is block storage, generally single-AZ and attached to one instance. EFS is a multi-AZ NFS file system shared by many instances. FSx provides managed Windows, Lustre or NetApp file systems.

27. How do you migrate large data to AWS?

Answer: DataSync is for online synchronization. Snowball is for physical transfer of TBs or PBs. S3 Transfer Acceleration improves global uploads over the internet. Choose based on bandwidth and data size.

28. S3 Cross-Region Replication vs. Same-Region Replication?

Answer: CRR supports DR and compliance. SRR supports log aggregation or cross-account access. Both require versioning.

PART 5: DATABASE

29. RDS vs. Aurora?

Answer: RDS manages traditional engines such as MySQL and PostgreSQL. Aurora is AWS-built and compatible with MySQL/PostgreSQL, with a separated storage layer and replication across AZs. Choose based on performance, compatibility, cost and operational needs.

30. RDS Multi-AZ vs. Read Replicas?

Answer: Multi-AZ provides synchronous standby for HA and failover, not read scaling. Read Replicas are asynchronous copies for read throughput. Use both when you need availability and performance.

31. When to use DynamoDB?

Answer: Use it for single-digit millisecond latency, key-value/document workloads and unpredictable scale. Avoid complex relational queries and large blobs. Use Global Tables for multi-region active-active where appropriate.

32. DynamoDB Partition Key Design?

Answer: Choose a high-cardinality key to avoid hot partitions. Use composite partition and sort keys for query flexibility, and GSIs for alternative access patterns.

33. ElastiCache Redis vs. Memcached?

Answer: Redis supports persistence, pub/sub, complex data structures and Multi-AZ HA. Memcached is simpler and multithreaded without persistence. Choose according to required features and operational model.

34. When to use Redshift?

Answer: Use it for petabyte-scale SQL data warehousing. Redshift Serverless suits variable workloads; Athena suits ad-hoc SQL over S3.

PART 6: SECURITY & IAM

35. IAM User vs. Role vs. Group?

Answer: A user has long-lived credentials and should generally be avoided for applications. A role provides temporary STS credentials for services and applications. A group is a collection of users. Prefer IAM Identity Center for humans and roles for workloads.

36. What is the Principle of Least Privilege?

Answer: Grant only required permissions. Use IAM Access Analyzer to find unused permissions, then tighten policies using CloudTrail evidence.

37. KMS vs. Secrets Manager vs. Parameter Store?

Answer: KMS manages encryption keys. Secrets Manager rotates database credentials and other secrets. Parameter Store provides hierarchical configuration storage. Use each for its intended purpose.

38. What is AWS WAF?

Answer: AWS WAF is a web application firewall that helps protect against common web exploits, bots and abuse through rules and rate limiting. It can attach to CloudFront, ALB or API Gateway.

39. GuardDuty vs. Security Hub vs. Inspector?

Answer: GuardDuty detects threats. Security Hub provides a central compliance and security dashboard. Inspector scans EC2 and container workloads for vulnerabilities.

40. How do you secure cross-account access?

Answer: Use IAM roles with sts:AssumeRole and External IDs where needed. Federate identity through IAM Identity Center and use SCPs to restrict dangerous actions.

PART 7: INTEGRATION & SERVERLESS

41. SQS vs. SNS vs. EventBridge vs. Kinesis?

Answer: SQS is a pull-based queue. SNS is pub/sub fan-out. EventBridge is an event bus for AWS and SaaS events. Kinesis is real-time streaming with replay. Use EventBridge for events and SQS for buffering.

42. SQS Standard vs. FIFO?

Answer: Standard provides at-least-once delivery, best-effort ordering and high throughput. FIFO provides ordering and deduplication with lower throughput. Use FIFO when ordering and exactly-once processing semantics are required.

43. Lambda Concurrency: Reserved vs. Provisioned?

Answer: Reserved concurrency caps maximum executions and protects downstream systems. Provisioned concurrency keeps environments warm for latency-sensitive APIs.

44. Step Functions vs. SQS-driven Lambda?

Answer: Step Functions provides visual workflows, retries, error handling and parallel branches. SQS plus Lambda is simpler and cheaper for asynchronous processing. Use Step Functions for complex orchestration.

PART 8: DESIGN SCENARIOS

45. Design a highly available web application.

Answer: CloudFront → ALB across multiple AZs → ASG with EC2/Fargate → Aurora Multi-AZ. Store static assets in S3, cache with ElastiCache.

46. Design a serverless REST API for 100K RPS.

Answer: API Gateway HTTP API → Lambda with Provisioned Concurrency → DynamoDB On-Demand. Cache hot paths at the edge or API layer and use X-Ray for tracing. Validate quotas and load-test before production.

47. Design a Data Lake.

Answer: S3 raw and curated zones → Glue Catalog → Athena for ad-hoc analysis → Redshift for BI. Use Lake Formation for governance.

48. Troubleshooting sudden slowness.

Answer: Check CloudWatch dashboards for ALB latency, database CPU and errors. Use X-Ray to identify bottlenecks and check throttling in DynamoDB or Lambda concurrency limits.

49. Migrating a monolith to AWS.

Answer: Use the 6 Rs. Often start with Rehost using Application Migration Service, then refactor gradually into containers using the Strangler Fig pattern. Use DMS for database migration.

50. Tell me about a system you designed.

Answer: Use the STAR method. Focus on trade-offs, such as consistency versus availability, explain why alternatives were rejected and mention relevant Well-Architected pillars.

PREPARATION TIPS

Focus on trade-offs. A junior answer says "Use S3 because it's cheap." A senior answer says "Use S3 Standard for 30 days, then lifecycle to Glacier Deep Archive to reduce cost while accepting the retrieval SLA."

Whiteboard a three-tier application, serverless API and multi-region DR setup from memory. Stay aware of current AWS services such as Amazon Q and Bedrock, and updates to Control Tower.

ENTERPRISE AGENTIC AI/RAG PLATFORM ON AWS FOR A FINANCIAL INSTITUTION

This production-grade architecture isolates compute inside private subnets, decouples execution logic from foundation-model inference and maintains strict security with human-in-the-loop oversight.

Edge and ingress: CloudFront, WAF and Shield intercept traffic at the edge; ALB terminates TLS, performs path-based routing and forwards clean traffic to private compute.

Core orchestration layer: An AI Gateway manages rate limiting, request validation, API-key rotation and telemetry. An Agent Orchestrator manages conversation state, task decomposition and step-by-step workflows. A RAG Service handles chunking, embeddings and vector retrieval.

Execution and data integration: Guardrails scan prompts and outputs for PII leakage, toxicity, hallucination risks and prompt injection. OpenSearch Serverless vector embeddings and PostgreSQL/pgvector provide hybrid retrieval. Tools may include risk and trade APIs, SQL and MCP interfaces.

Inference and HITL: Amazon Bedrock provides managed foundation-model inference. Human approval is a circuit breaker for high-risk actions such as executing trades or modifying databases.

GENERATIVE AI, BEDROCK, AMAZON Q AND RAG

1. What is Amazon Bedrock and how does it differ from SageMaker?

Answer: Bedrock is a fully managed service providing foundation models through a unified API for rapid integration. SageMaker is a broader ML platform for building, training and deploying custom models. Use Bedrock for managed foundation models and SageMaker for custom training or fine-tuning.

2. What are the key features of Amazon Bedrock Agents?

Answer: They execute multi-step tasks across company systems and data sources, break down requests, call APIs, manage conversation history and integrate with Knowledge Bases for RAG.

3. Explain Model Distillation in Bedrock.

Answer: Distillation trains a smaller model from a larger teacher model to reduce cost and latency for a specific use case while retaining useful accuracy.

4. How does Bedrock handle data privacy?

Answer: Bedrock inputs and outputs are not used by default to train the underlying foundation models. Data is encrypted at rest and in transit, and VPC endpoints can keep traffic within the AWS network.

5. On-Demand vs. Provisioned Throughput in Bedrock?

Answer: On-Demand charges per token with no commitment and suits variable workloads. Provisioned Throughput commits model capacity for consistent latency and throughput.

6. What are Guardrails in Amazon Bedrock?

Answer: Configurable safety filters that help prevent harmful or undesirable content. They can detect categories such as hate speech, sexual content and prompt injection, and can be applied across models.

7. How do you evaluate FM performance in Bedrock?

Answer: Use Bedrock Model Evaluation for automated metrics such as accuracy, robustness and toxicity, together with human evaluation workflows.

8. What is a Knowledge Base in Amazon Bedrock?

Answer: A managed RAG capability that connects models to private data sources, handling ingestion, chunking, embedding and vector storage.

9. What vector databases are supported natively by Bedrock Knowledge Bases?

Answer: Supported options include Aurora PostgreSQL, OpenSearch Serverless, Pinecone, Redis Enterprise Cloud and MongoDB Atlas. Select based on scale, latency, governance and query requirements.

10. How does Bedrock handle chunking in RAG?

Answer: Knowledge Bases offer fixed-size, hierarchical and semantic chunking. Custom chunking can be implemented when document structure needs specialized treatment.

11. What is the role of Amazon Titan Embeddings in RAG?

Answer: Titan Embeddings converts text into vectors stored in a vector database. Queries are embedded and compared with stored vectors to retrieve semantically relevant context.

12. How do you improve retrieval accuracy in RAG?

Answer: Use hybrid keyword and semantic search, re-ranking models and optimized chunking. Evaluate retrieval quality separately from generation quality.

13. Can Knowledge Bases connect to real-time data?

Answer: Yes. Agents can call Lambda functions or API connectors to fetch real-time information during execution, while Knowledge Bases are commonly used for indexed source data.

14. What is Amazon Q Developer?

Answer: An AI assistant for software development that integrates with IDEs to suggest code, generate tests, debug errors, explain code and support application modernization.

15. How does Amazon Q Business differ from Q Developer?

Answer: Q Business connects to enterprise repositories to answer business questions and summarize information. Q Developer focuses on coding and IT operations.

16. What is Agentic AI in AWS?

Answer: Systems that perceive, reason, act and iterate toward goals. AWS implementations can use Bedrock Agents and related services to plan workflows, call tools and respond to feedback.

17. How do you build a custom agent in Bedrock?

Answer: Define an Action Group through an OpenAPI schema, select a foundation model and optionally attach a Knowledge Base. The agent maps user intent to API calls and returns results.

18. What is the role of CodeCatalyst in Agentic AI?

Answer: Where available in an organization's toolchain, development assistants can generate IaC templates and application scaffolding from natural-language prompts. Validate current service availability and governance before standardizing.

19. How does Amazon Q handle enterprise security?

Answer: Q should respect IAM permissions and existing data-access controls so users receive only authorized information. Configure identity, connectors and data governance carefully.

20. What are Q Apps?

Answer: Low-code or no-code generative AI applications built on Q Business for team-specific assistants such as HR or sales workflows.

21. How do you deploy a Bedrock application to production?

Answer: Use SAM or CDK for repeatable infrastructure, expose APIs through API Gateway where appropriate, use CloudFront for global access, and enforce least-privilege IAM. Add evaluation, observability, guardrails and rollback procedures.

22. How do you monitor Bedrock API usage and costs?

Answer: Use CloudWatch for token counts, latency and errors; Cost Explorer and tags for spend attribution; and Budgets for alerts.

23. How do you log prompts and responses for auditing?

Answer: Enable model invocation logging to CloudWatch Logs or S3 where appropriate. Capture prompts, outputs and metadata only under approved data-retention and masking controls.

24. What is Amazon Bedrock Flows?

Answer: A visual way to chain models, Knowledge Bases, Agents and Lambda functions into a workflow, reducing custom orchestration code.

25. How do you test hallucinations in a RAG application?

Answer: Use faithfulness evaluation and grounding checks that compare the answer with retrieved documents. Flag claims unsupported by the context.

26. What is the role of Lambda in GenAI architectures?

Answer: Lambda preprocesses inputs, post-processes model outputs, calls external APIs and implements business logic such as controlled database writes.

27. How do you secure GenAI applications against prompt injection?

Answer: Combine Guardrails, input validation, separate system and user contexts, least-privilege tools, output validation and adversarial testing. Never rely on the model alone for authorization.

28. What is Human-in-the-Loop in Bedrock?

Answer: Route ambiguous or high-risk decisions to human reviewers before finalizing output or executing an action.

29. How do you manage versioning for Bedrock Agents?

Answer: Publish agent versions and associate aliases such as dev and prod. This supports controlled updates, rollback and testing.

30. What is the Well-Architected Lens for GenAI?

Answer: Guidance for GenAI workloads covering responsible AI, privacy, security, cost/token management, reliability and performance.

31. How do you handle rate limiting in Bedrock?

Answer: Respect service quotas, implement exponential backoff, manage concurrency in the application and request quota increases when justified.

32. What is the role of EventBridge in Agentic AI?

Answer: EventBridge can trigger Agents or Flows from events, such as a new S3 object triggering a summarization workflow.

EKS AND AI/GENAI

1. What is Amazon EKS and how does it differ from ECS?

Answer: EKS is managed Kubernetes for portability and complex Kubernetes ecosystems. ECS is AWS-native orchestration that is simpler and deeply integrated with AWS APIs. Choose based on ecosystem, portability and operating complexity.

2. Explain the EKS Control Plane vs. Data Plane.

Answer: AWS manages the highly available control plane. The data plane contains worker capacity on EC2, Fargate or Outposts where containers run; responsibility depends on the selected mode.

3. EKS on EC2 vs. EKS on Fargate?

Answer: EC2 node groups provide control, GPU support and lower cost at scale but require management. Fargate eliminates node management and charges by requested resources.

4. What is an EKS Add-on?

Answer: Managed cluster components such as VPC CNI, CoreDNS and kube-proxy, with lifecycle and compatibility support.

5. How does EKS handle version upgrades?

Answer: Upgrade the control plane, then node groups and add-ons according to the compatibility matrix. Use rolling or blue/green approaches for critical clusters.

6. What is EKS Auto Mode?

Answer: An automated operating mode that helps provision, scale and optimize node capacity, reducing manual node-group management. Confirm current capabilities and regional availability before adoption.

7. How does networking work in EKS?

Answer: The AWS VPC CNI assigns VPC IP addresses to pods, enabling them to communicate with AWS resources under VPC security controls. Plan subnet IP capacity carefully.

8. Security Groups for Pods vs. Node Security Groups?

Answer: Node security groups protect EC2 nodes. Pod security groups allow fine-grained rules for selected pods and stronger service isolation.

9. How do you secure access to the EKS API Server?

Answer: Use IAM and Kubernetes RBAC, restrict endpoint access, prefer private access where suitable and enable CloudTrail auditing. Use pod-level roles for AWS permissions.

10. What is IRSA?

Answer: IAM Roles for Service Accounts uses OIDC federation so Kubernetes service accounts assume least-privilege IAM roles without long-lived credentials in pods.

11. How do you expose an EKS application to the internet?

Answer: Use the AWS Load Balancer Controller to provision an ALB or NLB. ALB is preferred for HTTP/HTTPS path routing and WAF integration.

12. What is the AWS Load Balancer Controller?

Answer: A Kubernetes controller that manages AWS ALBs and NLBs through Kubernetes Ingress and Service definitions.

13. How do you provide persistent storage to EKS pods?

Answer: Use CSI drivers. EBS provides high-performance block storage; EFS provides shared multi-AZ file storage; FSx supports specialized Lustre or Windows workloads.

14. EBS vs. EFS in EKS?

Answer: EBS is single-AZ block storage with high IOPS and low latency, often for databases. EFS is multi-AZ shared storage for multiple pods.

15. How do you handle secrets in EKS?

Answer: Prefer Secrets Manager or Parameter Store with the Secrets Store CSI Driver, KMS encryption and rotation rather than placing sensitive values in etcd.

16. Why run GenAI workloads on EKS instead of SageMaker?

Answer: Use EKS for infrastructure control, custom Kubernetes operators or GPU inference servers such as vLLM/TGI. Use SageMaker for managed ML pipelines and model hosting.

17. How do you manage GPUs in EKS?

Answer: Install the NVIDIA device plugin, use GPU instance types and use Karpenter or Cluster Autoscaler to provision nodes. Verify drivers and AMI compatibility.

18. What is Karpenter and why use it?

Answer: Karpenter provisions nodes based on pod requirements, can support GPUs and Spot capacity, and can reduce waste compared with fixed node groups.

19. How do you optimize AI inference costs on EKS?

Answer: Use Spot for fault-tolerant workloads, batching through Triton or vLLM, right-sized nodes and scale-to-zero mechanisms such as KEDA where suitable.

20. Can EKS integrate with Bedrock?

Answer: Yes. EKS applications can call Bedrock through IAM roles such as IRSA, and EKS can host preprocessing, RAG orchestration or integration services.

21. What is KServe or Seldon Core?

Answer: Open-source Kubernetes model-serving frameworks supporting canary deployments, A/B testing and autoscaling for custom models.

22. How do you monitor EKS clusters?

Answer: Use CloudWatch Container Insights, ADOT, Prometheus and Grafana, including Amazon Managed Prometheus where appropriate.

23. How do you centralize EKS logs?

Answer: Run Fluent Bit to ship pod and node logs to CloudWatch or OpenSearch. For high-volume retention, stream through Firehose to S3.

24. What is X-Ray and how does it help in EKS?

Answer: X-Ray provides distributed tracing and helps identify latency bottlenecks across chains such as API Gateway, Lambda, EKS and Bedrock.

25. How do you debug CrashLoopBackOff?

Answer: Use `kubectl describe pod` for events and `kubectl logs` for application errors. Check missing configuration, secrets, startup failures and OOMKilled status, then review historical logs.

26. What is GitOps and how is it implemented in EKS?

Answer: Git is the source of truth and Argo CD or Flux synchronizes cluster state. Pull requests, policy checks and rollback provide auditability and reduce drift.

27. What is EKS Anywhere?

Answer: A way to run EKS-compatible clusters on premises or in other environments for hybrid requirements such as data residency and latency.

28. How do you handle multi-tenancy in EKS?

Answer: Use namespaces, resource quotas, limit ranges and network policies. For stronger isolation, use separate clusters or virtual clusters.

29. What is the role of App Mesh in EKS?

Answer: App Mesh provides service-to-service observability, traffic control and mTLS through sidecar proxies, without changing application code.

30. How do you secure the EKS software supply chain?

Answer: Scan images in ECR, sign them with Cosign and enforce admission policies with tools such as OPA Gatekeeper or Kyverno. Permit only trusted images.

SENIOR SOLUTIONS ARCHITECT — DEVELOPER TRANSFORMATION

1. What is the Strangler Fig Pattern and when do you recommend it?

Answer: Gradually replace parts of a monolith with microservices until the old system can be decommissioned. It is safer than a big-bang rewrite for large critical systems.

Case scenario: A Java monolith on EC2 with RDS Oracle handles authentication, catalog and orders. Extract Order Processing into API Gateway, Lambda and DynamoDB without disrupting the business.

Phase 1 — Facade: Put API Gateway in front of the monolith and initially proxy all routes to its ALB.

Phase 2 — New service: Build Lambda handlers, a DynamoDB bounded context, CodePipeline/CodeBuild/CodeDeploy and an anti-corruption layer.

Phase 3 — Data migration: Use DMS for historical migration and CDC. Prefer the new Lambda as the order system of record, publishing OrderCreated through EventBridge for legacy reporting synchronization.

Phase 4 — Traffic shift: Route order paths to Lambda, leave other routes on the monolith, use canary deployment and compare latency/errors with X-Ray and CloudWatch.

Phase 5 — Cleanup: Remove order code, archive old tables to S3 for compliance, then retire obsolete database structures.

2. How do you approach Mainframe Modernization on AWS?

Answer: Use AWS Mainframe Modernization or partner solutions. Choose Rehost, Refactor or Rebuild based on business risk, code complexity, skills and target outcomes. Decouple data with CDC where needed.

3. What is the role of AWS Application Migration Service?

Answer: MGN supports rehosting by replicating source servers at block level to AWS with minimal downtime. It is often a first step before refactoring and modernization.

4. How do you modernize Oracle or SQL Server to Aurora?

Answer: Use SCT for schema conversion and DMS for full load plus CDC. Run a dual-run or validation period, then cut over after data integrity and application behavior are verified.

5. What are the biggest cultural challenges in legacy modernization?

Answer: Resistance, limited cloud skills and fear of job loss. Address them with Centers of Excellence, hands-on training, transparent communication and measurable quick wins.

6. What is an Internal Developer Platform and why do enterprises need it?

Answer: An IDP is a self-service layer that abstracts infrastructure complexity, allowing developers to provision, deploy and monitor safely. Examples include Service Catalog, Backstage and Proton.

7. How does AWS Proton fit into Developer Transformation?

Answer: Proton provides standardized templates for infrastructure and pipelines so developers can deploy through approved, secure golden paths.

8. What metrics measure developer productivity?

Answer: DORA metrics: deployment frequency, lead time for changes, change failure rate and time to restore service.

9. How do you implement Golden Paths?

Answer: Create pre-approved, secure and documented templates for common architectures, making the compliant path the easiest path while retaining an exception process.

10. What is the role of Backstage in AWS environments?

Answer: Backstage is a developer portal that integrates with AWS tooling to discover APIs, create services and view documentation in one place.

11. CodePipeline vs. Jenkins/GitLab CI?

Answer: CodePipeline is managed and integrates natively with AWS. Jenkins and GitLab offer broader flexibility and ecosystems but require more operational ownership. Choose based on integration, governance and team capability.

12. How do you implement Blue/Green Deployments on AWS?

Answer: Use CodeDeploy and ALB for EC2/ECS, weighted Lambda aliases for Lambda, Argo Rollouts or Flagger for EKS, and Route 53 weighted routing where DNS-level control is appropriate.

13. What is GitOps and how do you enforce it?

Answer: Use Git as the source of truth, Argo CD or Flux for reconciliation, and mandatory pull requests, peer review and automated policy checks to prevent drift.

14. How do you secure CI/CD?

Answer: Generate SBOMs, scan ECR images, use least-privilege roles, sign artifacts with Cosign and run IaC and security checks such as Checkov or tfsec.

15. What is Shift Left security?

Answer: Integrate security in the IDE, commit and build stages rather than waiting until production. Early scanning reduces remediation cost and risk.

16. How do you handle a customer who says Cloud is too expensive?

Answer: Conduct a Well-Architected cost review, identify idle and over-provisioned resources, use tagging and budgets, and apply FinOps showback or chargeback. Often the issue is waste rather than cloud itself.

17. Describe a time you influenced a technical decision against a customer preference.

Answer: Use STAR. Explain the customer's original choice, quantify the risk or cost, build a proof of concept, compare alternatives and state the measurable result. Do not invent a result; use your actual evidence.

18. How do you assess cloud maturity?

Answer: Use the AWS Cloud Adoption Framework perspectives: Business, People, Governance, Platform, Security and Operations. Turn gaps into a prioritized roadmap.

19. What is the role of a Center of Excellence?

Answer: A cross-functional team defines standards, creates reusable assets and trains teams. It acts as an internal consultancy that scales cloud capability.

20. How do you align technical transformation with business goals?

Answer: Start with business KPIs, map initiatives directly to them and require each architectural recommendation to articulate measurable business value.

21. What is Event-Driven Architecture and why is it key to modernization?

Answer: EDA decouples producers and consumers through events, enabling legacy and modern services to coexist and allowing gradual decomposition.

22. How do you handle distributed transactions?

Answer: Avoid two-phase commits where possible. Use Saga choreography or orchestration, with Step Functions useful for retries and compensating actions.

23. What is an Anti-Corruption Layer?

Answer: A translation layer that prevents legacy models from leaking into a new architecture and translates requests and responses between systems.

24. How do you modernize .NET Framework applications?

Answer: Use App2Container, run containers on ECS or EKS, migrate suitable workloads to modern .NET and use Windows containers when full-framework compatibility is required.

25. What is the role of Amazon MQ?

Answer: Amazon MQ supports standard protocols such as JMS and AMQP when native SNS/SQS integration is not practical. It can be a stepping stone to native AWS services.

26. How does Amazon Q Developer accelerate transformation?

Answer: It can analyze legacy code, suggest modernization, generate tests and help translate code. Treat output as assisted engineering requiring review and testing.

27. How do you govern GenAI in an enterprise?

Answer: Establish an AI CoE, use Guardrails, federated identity, CloudTrail and cost tags, and define policies for privacy, retention, approved models, tools and human approval.

28. What is Agentic CI/CD?

Answer: AI agents can detect failed builds, analyze logs, suggest fixes and create pull requests, or recommend infrastructure changes. Keep approvals and least privilege for production actions.

29. How do you measure GenAI ROI in development?

Answer: Track code acceptance rate, reduction in boilerplate time, lead time, defect density, review effort and adoption, while measuring quality and security—not just output volume.

30. What is the biggest risk of GenAI in coding?

Answer: IP leakage and security vulnerabilities in generated code. Mitigate with enterprise controls, approved tools, code review, static analysis, data policies and developer training.

AWS MIGRATION SERVICES: ON-PREMISES OR OTHER CLOUDS TO AWS

Moving from on-premises or Azure/GCP to AWS is not one-size-fits-all. Use the 6 Rs: Rehost, Replatform, Refactor, Repurchase, Retire and Retain.

1. Discovery and planning

Migration Hub aggregates migration progress. Application Discovery Service maps servers and dependencies. Migration Evaluator uses usage data to recommend right-sized AWS capacity and estimate economics.

2. Data transfer

DataSync transfers active files online with incremental sync and verification. Snow Family supports offline transfer when network bandwidth is insufficient for TBs/PBs. S3 Transfer Acceleration improves long-distance uploads.

3. Server and VM migration

AWS Application Migration Service installs a lightweight agent on VMware, Hyper-V, Azure, GCP or physical servers and continuously replicates blocks to AWS. Test instances can be launched repeatedly; at cutover, replicate the final delta and launch the target.

4. Database migration

DMS uses a replication instance for full load and CDC while the source remains online. SCT converts schemas and database code between engines. This supports migrations such as Oracle or SQL Server to Aurora, subject to compatibility testing.

5. Mainframe and legacy modernization

AWS Mainframe Modernization and partner solutions provide managed runtimes or conversion paths for COBOL and PL/I. Select rehost, replatform, refactor or rebuild according to risk and business value.

6. Application and code modernization

App2Container analyzes Java or .NET applications and generates container artifacts for ECS or EKS. Amazon Q Developer can assist with code analysis, refactoring, tests and translation, with human review.

7. Post-migration cleanup

Use Cost Explorer and Trusted Advisor to identify unused resources and avoid paying for both source and target longer than necessary.

6 Rs framework:

Rehost — MGN — move quickly with minimal code change.

Replatform — RDS, Elastic Beanstalk or ECS — adopt managed services with limited changes.

Refactor — App2Container, Lambda, EKS or Q Developer — modernize architecture and code.

Repurchase — SaaS such as Salesforce or Workday — replace custom applications.

Retire — Cost Explorer — turn off unused systems.

Retain — hybrid connectivity such as Direct Connect — keep workloads on premises temporarily.

Interview answer: “How would you migrate a critical Oracle database and Java monolith from on premises to AWS with minimal downtime?”

“I would use a hybrid strategy focused on minimal downtime and risk mitigation:

1. Discovery: use Application Discovery Service to map application and database dependencies.

2. Database migration: use SCT to assess and convert the Oracle schema to Aurora PostgreSQL, then DMS for full load and CDC. Keep both systems synchronized while validating.

3. Application migration: use MGN to replicate the Java VM and repeatedly test it in AWS.

4. Cutover: stop writes to Oracle, let DMS flush final CDC changes, update DNS or the AWS ALB path, and start the validated AWS application.

5. Future state: use App2Container or Q Developer to gradually refactor the monolith into microservices on ECS or EKS using the Strangler Fig pattern.”